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REFRIGERATOR CAN STORAGE ORGANIZATION

Abstract

A beverage chute is configured to removably mount within a refrigerator appliance. The beverage chute includes a front wall, a back wall, a first side wall, a second side wall, and a floor. The floor of the beverage chute is angled downward in the vertical direction from the back wall to the front wall.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to organizational systems within refrigerator appliances, for example racks and shelves.

BACKGROUND OF THE INVENTION

[0002] The primary function of a refrigerator is to store perishable food items, such as meat, fish,

vegetables, fruits, and beverages, to allow the perishable food items to maintain a fresh state. Generally, a refrigerator includes a refrigeration system cooling one or more storage chambers, such as a freezing chamber and a fresh-food chamber. Doors are disposed at one or more sides of the refrigerator body and are configured to open and close the storage chambers. In order to effectively utilize the storage chambers inside the refrigerator, numerous shelves are typically mounted inside of the storage chambers so as to accommodate storage of items thereon.

[0003] A door storage, such as a basket having storage products kept therein, can be disposed on a rear surface of the door. In the refrigerator, a large quantity of storage products can be kept in the storage of the storage chamber. As the depth of the storage chamber increases, a larger quantity of storage products may be kept in the storage of the storage chamber. A user may keep products in the storage chamber and the door storage according to the sizes or kinds of products. Due to various sizes and shapes of objects being put into the storage chamber, unusable space may be created in the storage chambers, often referred to as dead space.

[0004] Accordingly, a refrigerator appliance with systems for making use of the dead space within the refrigerator appliance would be advantageous.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

[0006] In one example embodiment, a refrigerator appliance is provided. The refrigerator appliance defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The refrigerator appliance includes a cabinet defining a food storage chamber. The food storage chamber extends between a top portion and a bottom portion along the vertical direction, a first side portion and a second side portion along the lateral direction, and a front portion and a back portion along the transverse direction. A first support member and a second support member are disposed proximate the back portion of the food storage chamber for providing a plurality of shelf mounting positions. A shelf extends between and mounted on the first and second support members within the fresh food chamber. The shelf extends between a front side and a back side. A beverage chute is positioned under the shelf. The beverage chute is mounted between the front side of the shelf and the first or second support members. The beverage chute includes a front wall, a back wall, a first side wall, a second side wall, and a floor.

[0007] In another example embodiment, a beverage chute is provided. The beverage chute defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The beverage chute is configured to removably mount within a refrigerator appliance. The beverage chute includes a front wall, a back wall, a first side wall, a second side wall, and a floor. The floor of the beverage chute is angled downward in the vertical direction from the back wall to the front wall.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 is a perspective view of a refrigerator appliance according to an example embodiment of the present disclosure.

[0011] FIG. 2 is a perspective view of the example refrigerator appliance of FIG. 1 with doors shown in an open configuration.

[0012] FIG. 3 is a perspective view of a beverage chute shown in FIG. 2 and according to an example embodiment of the present disclosure.

[0013] FIG. 4 is side, elevation view of the beverage chute of FIG. 3 according to an example embodiment of the present disclosure.

[0014] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0016] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “upstream” and “downstream” refer to the relative flow direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the flow direction from which the fluid flows, and “downstream” refers to the flow direction to which the fluid flows. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”).

[0017] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a ten percent (10%) margin.

[0018] Referring now to the figures, an example appliance will be described in accordance with example aspects of the present subject matter. Specifically, FIG. 1 provides a perspective view of an example refrigerator appliance **100**, and FIG. 2 illustrates refrigerator appliance **100** with some of the doors in the open position. As illustrated, refrigerator appliance **100** generally defines a vertical direction V, a lateral direction L, and a transverse direction T that are mutually perpendicular and form an orthogonal coordinate system.

[0019] According to example embodiments, refrigerator appliance **100** includes a cabinet **102** that is generally configured for containing and/or supporting various components of refrigerator appliance **100** and which may also define one or more internal chambers or compartments of refrigerator appliance **100**. In this regard, as used herein, the terms “cabinet,” “housing,” and the like are generally intended to refer to an outer frame or support structure for refrigerator appliance **100**, e.g., including any suitable number, type, and configuration of support structures formed from any suitable materials, such as a system of elongated support members, a plurality of interconnected panels, or some combination thereof. It should be appreciated that cabinet **102** does not necessarily require an enclosure and may simply include open structure supporting various elements of refrigerator appliance **100**. By contrast, cabinet **102** may enclose some or all portions

of an interior of cabinet **102**. It should be appreciated that cabinet **102** may have any suitable size, shape, and configuration while remaining within the scope of the present subject matter.

[0020] As illustrated, cabinet **102** generally extends between a top portion **104** and a bottom portion **106** along the vertical direction V, between a first side portion **108** (e.g., the left side when viewed from the front as in FIG. 1) and a second side portion **110** (e.g., the right side when viewed from the front as in FIG. 1) along the lateral direction L, and between a front portion **112** and a back portion **114** along the transverse direction T. In general, terms such as “left,” “right,” “front,” “rear,” “top,” or “bottom” are used with reference to the perspective of a user accessing refrigerator appliance **100**. As shown in FIG. 2, cabinet **102** may include a top wall **160** and a pair of side walls **162**, e.g., that are spaced apart along the lateral direction L.

[0021] Cabinet **102** defines chilled chambers for receipt of food items for storage. In particular, cabinet **102** defines fresh food chamber **122** positioned at or adjacent top portion **104** of cabinet **102** and a freezer chamber **124** arranged at or adjacent bottom portion **106** of cabinet **102**. As such, refrigerator appliance **100** is generally referred to as a bottom mount refrigerator. It is recognized, however, that the benefits of the present disclosure apply to other types and styles of refrigerator appliances such as, e.g., a top mount refrigerator appliance, a side-by-side style refrigerator appliance, or a single door refrigerator appliance. Consequently, the description set forth herein is for illustrative purposes only and is not intended to be limiting in any aspect to any particular appliance or configuration.

[0022] Refrigerator doors **128** are rotatably hinged to an edge of cabinet **102** for selectively accessing fresh food chamber **122**. In addition, a freezer door **130** is arranged below refrigerator doors **128** for selectively accessing freezer chamber **124**. Freezer door **130** is coupled to a freezer drawer (not shown) slidably mounted within freezer chamber **124**. In general, refrigerator doors **128** form a seal over a front opening **132** defined by cabinet **102**. In this regard, a user may place items within fresh food chamber **122** through front opening **132** when refrigerator doors **128** are open and may then close refrigerator doors **128** to facilitate climate control. Refrigerator doors **128** and freezer door **130** are shown in the closed configuration in FIG. 1. One skilled in the art will appreciate that other chamber and door configurations are possible and within the scope of the present invention.

[0023] FIG. 2 provides a perspective view of refrigerator appliance **100** shown with refrigerator doors **128** in the open position. As shown in FIG. 2, various storage components are mounted within fresh food chamber **122** to facilitate storage of food items therein as will be understood by those skilled in the art. In particular, the storage components may include pans **134** and shelves **136**. Each of these storage components are configured for receipt of food items (e.g., beverages and/or solid food items) and may assist with organizing such food items. As illustrated, pans **134** may be mounted on refrigerator doors **128**, referred to then as bins, or may slide into a receiving space in fresh food chamber **122**. It should be appreciated that the illustrated storage components are used only for the purpose of explanation and that other storage components may be used and may have different sizes, shapes, and configurations.

[0024] Turning back to FIG. 1, a dispensing assembly **140** is generally configured for dispensing liquid water and/or ice. Dispensing assembly **140** and its various components may be positioned at least in part within a dispenser recess **142** defined on one of refrigerator doors **128**. In this regard, dispenser recess **142** is defined on a front portion **112** of refrigerator appliance **100** such that a user may operate dispensing assembly **140** without opening refrigerator door **128**. In addition, dispenser recess **142** is positioned at a predetermined elevation convenient for a user to access ice and enabling the user to access ice without the need to bend-over. In the exemplary embodiment, dispenser recess **142** is positioned at a level that approximates the chest level of a user.

[0025] Dispensing assembly **140** includes an ice dispenser **144** including a discharging outlet **146** for discharging ice from dispensing assembly **140**. An actuating mechanism **148**, shown as a paddle, is mounted below discharging outlet **146** for operating ice or water dispenser **144**. In

alternative exemplary embodiments, any suitable actuating mechanism may be used to operate ice dispenser **144**. For example, ice dispenser **144** can include a sensor (such as an ultrasonic sensor) or a button rather than the paddle. Discharging outlet **146** and actuating mechanism **148** are an external part of ice dispenser **144** and are mounted in dispenser recess **142**. By contrast, refrigerator door **128** may define an icebox compartment **150** (FIG. 2) housing an icemaker and an ice storage bin (not shown) that are configured to supply ice to dispenser recess **142**.

[0026] A control panel **152** is provided for controlling the mode of operation. For example, control panel **152** includes one or more selector inputs **154**, such as knobs, buttons, touchscreen interfaces, etc., such as a water dispensing button and an ice-dispensing button, for selecting a desired mode of operation such as crushed or non-crushed ice. In addition, inputs **154** may be used to specify a fill volume or method of operating dispensing assembly **140**. In this regard, inputs **154** may be in communication with a processing device or controller **156**. Signals generated in controller **156** operate refrigerator appliance **100** and dispensing assembly **140** in response to selector inputs **154**. Additionally, a display **158**, such as an indicator light or a screen, may be provided on control panel **152**. Display **158** may be in communication with controller **156**, and may display information in response to signals from controller **156**.

[0027] As used herein, “processing device” or “controller” may refer to one or more microprocessors or semiconductor devices and is not restricted necessarily to a single element. The processing device can be programmed to operate refrigerator appliance **100**, dispensing assembly **140** and other components of refrigerator appliance **100**. The processing device may include, or be associated with, one or more memory elements (e.g., non-transitory storage media). In some such embodiments, the memory elements include electrically erasable, programmable read only memory (EEPROM). Generally, the memory elements can store information accessible by a processing device, including instructions that can be executed by processing device. Optionally, the instructions can be software or any set of instructions and/or data that when executed by the processing device, cause the processing device to perform operations.

[0028] Referring again to FIG. 2, for this example embodiment, fresh food chamber **122** of refrigerator appliance **100** includes a first support member **170** and a second support member **172** disposed proximate the back portion **114**, e.g., mounted to a back wall **115** of cabinet **120**. The first and second support members **170** and **172** may be oriented generally along the vertical direction V. One or more adjustable shelves **136** may be mounted to the first and second support members **170** and **172**. In general, adjustable shelves **136** may extend between a front side **137** and a back side **139** in the transverse direction T within refrigerator appliance **100**. First and second support members **170** and **172** may structurally support one or more adjustable shelves **136**. Moreover, first and second support members **170** and **172** may structurally support the weight of other components, e.g., the additional weight from a beverage chute **200** (FIG. 3). First and second support members **170** and **172** may be made of any suitable structural material, e.g., in some embodiments, first and second support members **170** and **172** may be made of steel.

[0029] In the embodiment illustrated in FIG. 2, four (4) adjustable shelves **136** are mounted within fresh food chamber **122** and are arranged in two columns and two rows as shown. Adjustable shelves **136** may be selectively positioned by a user in different shelf mounting positions within fresh food chamber **122**. For instance, one adjustable shelf **136** could be removed from its position and moved upward or downward along the vertical direction V or moved from a position proximate first side portion **108** to a position proximate second side portion **110** of refrigerator appliance **100** along the lateral direction L. Adjustable shelves **136** may also be removed from refrigerator appliance **100**. For example, if storage room is needed for a particularly tall pot, adjustable shelves **136** may be removed from refrigerator appliance **100** and stowed elsewhere. Although four (4) adjustable shelves **136** are depicted in FIG. 2, more or less than four (4) adjustable shelves **136** may be provided in refrigerator appliance **100**.

[0030] Turning now to FIGS. 3-4, illustrated in FIG. 3 is a perspective view of an example

embodiment of a beverage chute **200**, and illustrated in FIG. 4 is an elevation, side view of the example embodiment of beverage chute **200**. Beverage chute **200** may be generally defined by a front wall **202**, a back wall **204**, a first side wall **206**, a second side wall **208**, and a floor **210**. In general, beverage chute **200** may be formed as a single piece construction, e.g., beverage chute **200** may be fabricated through injection molding or additive manufacturing. In other example embodiments, beverage chute **200** may be a wire rack, e.g., a grid pattern of wires, bent to form the features of beverage chute **200** as will be described hereinbelow. In the illustrated embodiment shown in FIGS. 3-4, floor **210** may curve upward, in the vertical direction V, into the front wall **202** and the back wall **204**. A person having ordinary skill in the art would understand that such curvature of floor **210** is provided for example purposes only and may be, in certain example embodiments, omitted, resulting in a box-like structure at junctions between both front and back wall **202**, **204** and floor **210**.

[0031] In general, beverage chute **200** may be positioned under one of the adjustable shelves **136**, e.g., beverage chute **200** may be mounted between front side **137** and back side **139** of one of the adjustable shelves **136** and/or the first or second support members **170**, **172**. For example, beverage chute **200** may include a first mounting lip **212** extending backward along the transverse direction from front wall **202**. The first mounting lip **212** may be generally configured to removably couple to front side **137** of one of the adjustable shelves **136**. Additionally, a second mounting lip **214** may extend forward from the back wall **204** along the transverse direction of beverage chute **200**. The second mounting lip **214** may be generally configured to removably couple to one or more of the first or second support members **170**, **172** and/or back side **139** of one of the adjustable shelves **136**.

[0032] As may be seen in FIG. 4, floor **210** of beverage chute **200** may be angled downward, e.g., in the vertical direction V, from back wall **204** to front wall **202**, such that the floor **210** at back wall **204** is higher in the vertical direction V than the floor **210** at front wall **202**. Specifically, floor **210** may be angled at an angle α from a plane **216** defined by the lateral direction L and transverse direction T, where angle α may be between one degree (1°) and thirty-five degrees (35°), such as between two degrees (2°) and thirty degrees (30°), such as between five degrees (5°) and twenty degrees (20°) from plane **216**. Accordingly, floor **210** may generally define a slope downwards in the vertical direction V from back wall **204** to front wall **202**. As stated above, floor **210** may curve upward, in the vertical direction V, into the front wall **202** and the back wall **204**. The portions of curved floor **210** may deviate from angle α , however, floor **210** may maintain the general slope downwards in the vertical direction V from back wall **204** to front wall **202**.

[0033] Referring again to FIG. 3-4, in the present example embodiment, beverage chute **200** includes an inlet **220** defined in the first side wall **206**. In general, inlet **220** may be configured for receipt of beverage canisters **300**, e.g., soda/pop cans, energy drinks, adult beverages (alcoholic drinks), bottles, or other suitable beverage canisters, etc., such as a beverage canister having an internal volume of about twelve fluid ounces (12 fl. oz) or less. In particular, inlet **220** may define an opening **221** through first side wall **206**. Opening **221** may generally be circular shaped with a diameter between two and three quarters inches (2.75 in) and four inches (4 in), such as between three inches (3 in) and three and three quarters inches (3.75 in), such as between three and one quarter inches (3.25 in) and three and a half inches (3.5 in). In general, opening **221** of inlet **220** may be no less than two and six tenths inches (2.6 in), e.g., the diameter of the widest part of a typical soda beverage canister **300**. While opening **221** of inlet **220** is described and shown as circular, opening **221** may be any shape suitable for receiving beverage canisters **300**, such as square and rectangular shapes.

[0034] In the present example embodiment, beverage chute **200** may also include an outlet **222** defined in the first side wall **206**. In general, outlet **222** may be configured for dispensing beverage canisters **300**. In particular, outlet **222** may define an opening **223** through first side wall **206**. Opening **223** may generally be circular shaped with a diameter between three inches (3 in) and six

inches (6 in), such as between four inches (4 in) and five inches (5 in), such as approximately four and a half inches (4.5 in). In the present example embodiment, opening **223** of outlet **222** is illustrated as a rotated tear shape, e.g., having a circular form at one side and tapering to a point at the other side, or may taper to an arcuate end having a smaller radius than the circular form at the one side, such as a radius of about one half of the radius of the circular form at the one side. While illustrated having the teardrop shape, opening **223** of outlet **222** may have any shape suitable for a user to retrieve beverage canister **300**, e.g., reach into outlet **222**, grasp beverage canister **300**, and pull the beverage canister **300** out of beverage chute **200**.

[0035] As shown in FIG. 4, inlet **220** may be positioned on the first side wall **206** proximate the back wall **204** in the transverse direction T, and outlet **222** may be positioned on the first side wall **206** proximate the front wall **202** in the transverse direction T. Accordingly, as beverage canisters **300** are loaded into inlet **220**, gravity may advantageously roll the beverage canisters **300** down the slope (angle α) of floor **210** towards outlet **222**. While inlet **220** and outlet **222** are illustrated and described on first side wall **206**, a person having ordinary skill in the art would understand that in other example embodiments, inlet **220** and outlet **222**, and thus respective openings **221**, **223**, may be positioned on second side wall **208** as well as or instead of the first side wall **206**.

[0036] As shown in FIG. 4, beverage chute **200** may hold six (6) beverage canisters **300**. In general, beverage chute **200** may be sized along the transverse direction T to match a depth, e.g., from front side **137** to back side **139**, of the adjustable shelves **136**. However, beverage chute **200** may be larger or smaller in other example embodiments, depending on the depth of the shelf that beverage chute **200** is mounted thereto. Accordingly, depending on the length of beverage chute **200** in the transverse direction T, beverage chute **200** may hold between two (2) beverage canisters **300** and sixteen (16) beverage canisters **300**, such as between three (3) beverage canisters **300** and twelve (12) beverage canisters **300**, such as between four (4) beverage canisters **300** and eight (8) beverage canisters **300**. As such, beverage chute **200** may advantageously aid utilizing the dead space within refrigerator appliance **100** by holding beverage canisters **300** which would typically be placed on one or more of the adjustable shelves **136**. Additionally, since the canisters which would typically be placed on one or more of the adjustable shelves **136** are stored in beverage chute **200**, the emptied space on the adjustable shelves **136** advantageously becomes available for other items to be placed thereupon.

[0037] As may be seen from the above, a refrigerator appliance may include a drink, beverage, can, bottle storage chute that may use the dead space within the refrigerator appliance by attaching to an underside of a refrigerator shelf. The chute may include a forward slope, e.g., slopes downward from a rear side of the shelf towards a front side of the shelf. The beverage containers may be inserted into the chute at a rear portion of the chute, and gravity may roll the containers forward for access to a user of the refrigerator appliance. Based on a depth of the shelf, the chute may store varying number of cans.

[0038] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the

refrigerator appliance comprising: a cabinet defining a food storage chamber, the food storage chamber extending between a top portion and a bottom portion along the vertical direction, a first side portion and a second side portion along the lateral direction, and a front portion and a back portion along the transverse direction; a first support member and a second support member disposed proximate the back portion of the food storage chamber for providing a plurality of shelf mounting positions; a shelf extending between and mounted on the first and second support members within the food storage chamber, the shelf extending between a front side and a back side; and a beverage chute positioned under the shelf, the beverage chute mounted between the front side of the shelf and the first or second support members, the beverage chute comprising a front wall, a back wall, a first side wall, a second side wall, and a floor.

2. The refrigerator as in claim 1, wherein the beverage chute comprises a first mounting lip extending from the front wall, the first mounting lip removably coupled to the front side of the shelf.
3. The refrigerator as in claim 2, wherein the beverage chute comprises a second mounting lip extending from the back wall, the second mounting lip removably coupled to one or more of the first or second support members and the back side of the shelf.
4. The refrigerator as in claim 1, wherein the floor of the beverage chute is angled downward in the vertical direction from the back wall to the front wall.
5. The refrigerator as in claim 4, wherein the beverage chute comprises an inlet defined in the first side wall, the inlet configured for receipt of beverage canisters.
6. The refrigerator as in claim 5, wherein the inlet defines an opening between two and three quarters inches and four inches.
7. The refrigerator as in claim 5, wherein the beverage chute comprises an outlet defined in the first side wall, the outlet configured for dispensing beverage canisters.
8. The refrigerator as in claim 7, wherein the outlet defines an opening between three inches and six inches.
9. The refrigerator as in claim 7, wherein the inlet is positioned on the first side wall proximate the back wall in the transverse direction, and the outlet is positioned on the first side wall proximate the front wall in the transverse direction.
10. The refrigerator as in claim 4, wherein the floor is angled between one and thirty-five degrees from a plane defined by the lateral and transverse directions.
11. A beverage chute defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the beverage chute configured to removably mount within a refrigerator appliance, the beverage chute comprising: a front wall, a back wall, a first side wall, a second side wall, and a floor, wherein the floor of the beverage chute is angled downward in the vertical direction from the back wall to the front wall.
12. The beverage chute as in claim 11, wherein the beverage chute comprises a first mounting lip extending from the front wall, the first mounting lip removably coupled to a shelf within the refrigerator appliance.
13. The beverage chute as in claim 12, wherein the beverage chute comprises a second mounting lip extending from the back wall, the second mounting lip removably coupled to the shelf within the refrigerator appliance.
14. The beverage chute as in claim 11, wherein the beverage chute comprises an inlet defined in the first side wall, the inlet configured for receipt of beverage canisters.
15. The beverage chute as in claim 14, wherein the inlet defines an opening between two and three quarters inches and four inches.
16. The beverage chute as in claim 14, wherein the beverage chute comprises an outlet defined in the first side wall, the outlet configured for dispensing beverage canisters.
17. The beverage chute as in claim 16, wherein the outlet defines an opening between three inches and six inches.

18. The beverage chute as in claim 16, wherein the inlet is positioned on the first side wall proximate the back wall in the transverse direction, and the outlet is positioned on the first side wall proximate the front wall in the transverse direction.

19. The beverage chute as in claim 11, wherein the floor is angled between one and thirty-five degrees from a plane defined by the lateral and transverse directions.
